Rates and Unit Rates

A unit rate is how much for 1.

12 apples cost $6. How much for 1?

apples	12	6	2	1
dollars	6	3	1	0.5

Halve both, and halve again, until there is 1 apple.
$6 ÷ 12 = $0.50 each.

Per means “for each 1”

150 miles in 3 hours = 50 miles per hour

150 ÷ 3 = 50. Then 5 hours is 5 × 50 = 250 miles.

The better buy has the lower price for 1

4 pens for $2 is $0.50 each. 6 pens for $2.40 is $0.40 each: the better buy.

Your turn!

96 words in 4 minutes = ☐ words per minute

Rates and Unit Rates: Practice

Find 1 first. Then multiply.

Part 1 Find 1 in the table Divide down to 1, then multiply.

1)

tickets	4			
dollars	32			

2)

hours	3			
miles	165			

3)

boxes	5			
cookies	60			

4)

days	6			
pages	90			

5)

bags	2			
apples	14			

Part 2 Unit rates No help this time.

6) **120 miles in 3 hours = ____ miles per hour**

7) **84 words in 4 minutes = ____ words per minute**

Rates and Unit Rates: Practice (continued)

Part 2 Unit rates (continued)

8) 90 pages in 5 days = ☐ pages per day

9) 240 miles in 6 hours = ☐ miles per hour

10) 56 laps in 8 hours = ☐ laps per hour

11) 135 points in 9 games = ☐ points per game

12) 72 cookies in 3 batches = ☐ cookies per batch

13) 325 miles in 5 hours = ☐ miles per hour

14) 96 dollars in 12 hours = ☐ dollars per hour

15) 63 beads in 7 bracelets = ☐ beads per bracelet

16) 180 steps in 4 minutes = ☐ steps per minute

17) 144 dollars in 6 days = ☐ dollars per day

18) 85 words in 5 minutes = ☐ words per minute

19) 210 miles in 7 hours = ☐ miles per hour

Rates and Unit Rates: Practice (continued)

Part 2 Unit rates (continued)

20) 48 eggs in 4 boxes = ☐ eggs per box

21) 156 points in 3 games = ☐ points per game

Part 3 Price for 1 No help this time.

22) 6 pencils for $3.00 = ☐ each

23) 4 apples for $3.00 = ☐ each

24) 8 stickers for $2.00 = ☐ each

25) 5 notebooks for $6.00 = ☐ each

26) 3 pears for $1.35 = ☐ each

27) 10 erasers for $3.50 = ☐ each

28) 12 candies for $1.80 = ☐ each

29) 4 juices for $5.20 = ☐ each

30) 6 bagels for $5.10 = ☐ each

31) 5 cards for $3.00 = ☐ each

Rates and Unit Rates: Practice (continued)

Part 3 Price for 1 (continued)

32) 8 pens for $3.20 = ☐ each

33) 3 books for $6.75 = ☐ each

Part 4 Use the rate No help this time.

34) 45 miles per hour: 6 hours = ☐ miles

35) 12 pages per day: 9 days = ☐ pages

36) 8 dollars per hour: 15 hours = ☐ dollars

37) 65 words per minute: 4 minutes = ☐ words

38) 24 cookies per batch: 7 batches = ☐ cookies

39) 15 laps per hour: 8 hours = ☐ laps

40) 35 miles per hour: 12 hours = ☐ miles

41) 9 beads per bracelet: 14 bracelets = ☐ beads

42) 18 points per game: 11 games = ☐ points

43) 55 steps per minute: 6 minutes = ☐ steps

Rates and Unit Rates: Practice (continued)

Part 4 Use the rate (continued)

44) 14 dollars per day: 21 days = ☐ dollars

45) 6 eggs per box: 25 boxes = ☐ eggs

Part 5 More unit rates

46) 270 pages in 5 days = ☐ pages per day

47) 105 pages in 7 days = ☐ pages per day

48) 44 words in 4 minutes = ☐ words per minute

49) 504 words in 9 minutes = ☐ words per minute

50) 18 words in 2 minutes = ☐ words per minute

51) 112 dollars in 2 hours = ☐ dollars per hour

52) 140 dollars in 5 hours = ☐ dollars per hour

53) 354 words in 6 minutes = ☐ words per minute

54) 153 points in 3 games = ☐ points per game

Rates and Unit Rates: Practice (continued)

Part 5 More unit rates (continued)

55) 531 words in 9 minutes = ☐ words per minute

56) 77 pages in 7 days = ☐ pages per day

57) 192 miles in 3 hours = ☐ miles per hour

58) 90 dollars in 2 hours = ☐ dollars per hour

59) 120 pages in 8 days = ☐ pages per day

60) 213 miles in 3 hours = ☐ miles per hour

61) 130 words in 5 minutes = ☐ words per minute

62) 360 words in 8 minutes = ☐ words per minute

63) 54 pages in 6 days = ☐ pages per day

Part 6 More prices for 1

64) 9 bagels for $25.20 = ☐ each

65) 3 pens for $2.55 = ☐ each

Rates and Unit Rates: Practice (continued)

Part 6 More prices for 1 (continued)

66) 9 cards for $11.25 = ____ each

67) 12 bagels for $8.40 = ____ each

68) 5 pens for $4.25 = ____ each

69) 9 juices for $4.50 = ____ each

70) 8 cards for $1.20 = ____ each

71) 2 erasers for $1.80 = ____ each

72) 4 cards for $2.60 = ____ each

73) 6 bagels for $2.70 = ____ each

74) 5 bagels for $3.00 = ____ each

75) 2 juices for $4.60 = ____ each

Rates and Unit Rates: Practice (continued)

Part 7 Remember? Problems from earlier packets.

76) $11 : 4 = \square : 16$

77) $2.4 + 3.65 = \square$

78) $\frac{1}{2} \times \frac{3}{5} = \frac{\square}{\square}$

79) $6 \times 2.35 = \square$

80) $5.6 \div 7 = \square$

81) $12 + 6 \times 3 = \square$

Part 8 Word problems Find 1 first.

82) A car goes 270 miles in 6 hours. How far does it go in 1 hour?

Number sentence _____

Answer ☐ miles

Draw or show your work

83) A pack of 8 juice boxes costs $4. How much is 1 juice box?

Number sentence _____

Answer ☐

Draw or show your work

84) Mia types 42 words a minute. How many words can she type in 9 minutes?

Number sentence _____

Answer ☐ words

Draw or show your work

Rates and Unit Rates: Practice (continued)

Part 8 Word problems (continued)

85) Store A: 5 pens for $3. Store B: 4 pens for $2. Which store has the lower price for 1 pen?

Number sentence _____

Answer ☐

Draw or show your work

86) Challenge: A printer prints 36 pages in 3 minutes. How long does it take to print 180 pages?

Number sentence _____

Answer ☐ minutes

Draw or show your work

Check-In: Rates and Unit Rates

Show what you know. No help this time. Score: _____ / 8

Part 1 Find 1 in the table

1)

boxes	3			
pens	36			

Part 2 Unit rates

2) 64 words in 4 minutes = ____ words per minute

3) 108 points in 9 games = ____ points per game

4) 98 dollars in 7 days = ____ dollars per day

Part 3 Price for 1

5) 5 pens for $2.00 = ____ each

6) 6 cards for $1.80 = ____ each

Part 4 Use the rate

7) 55 miles per hour: 4 hours = ____ miles

8) 16 pages per day: 7 days = ____ pages